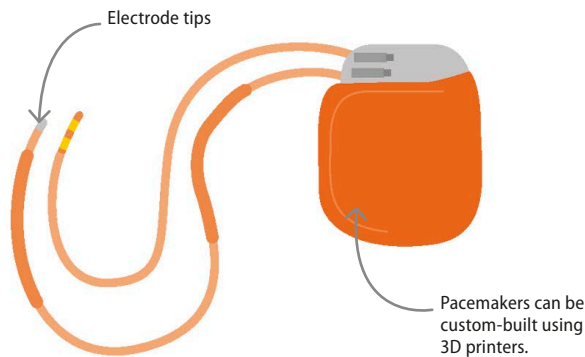


Implants

An implant is any device inserted inside the body. Some implants double as prosthetics, while others are enhancements, such as orthopaedic pins or rods that support damaged bones. Other implants monitor bodily functions and automatically administer medicine based on its readings. While implant surgery is becoming safer and more common, there's always a risk that the body will reject the implant.



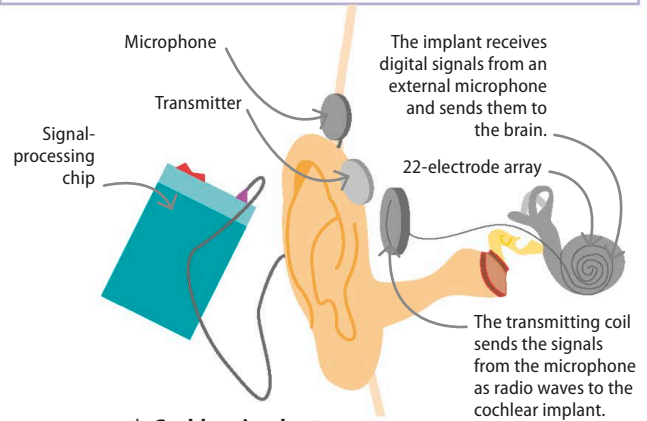
△ Pacemaker

A pacemaker consists of a battery, a generator, and wires with electrode tips implanted inside the chest cavity. It uses electric pulses to keep the heart beating in a regular rhythm. Certain pacemakers can also monitor a patient's vital signs, such as blood temperature, natural electrical activity, and breathing rate.

IN DEPTH

Going forward

It's impossible to tell where the future of biological interfaces is heading, but it's clear that they are becoming more common, and 3D printing is making them cheaper. There are many upcoming possibilities, from implanted chips used for identification, to sensors that heighten our senses or give us new ones.



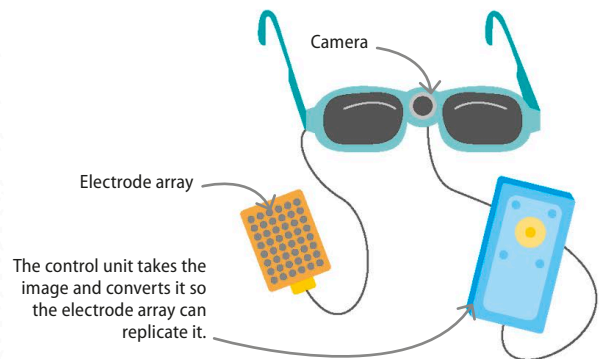
△ Cochlear implants

These implants are used for patients who are deaf or near-deaf. Unlike standard hearing aids, they require an implant attached to the cochlea, inside the ear canal. After surgery, it takes a few weeks of training for people with cochlear implants to interpret the new sounds they're hearing.



△ Eye implants

Companies are developing bionic eyes for people who are almost completely blind. Electrodes are implanted on and inside the eye in order to bypass natural photoreceptors and send electric signals directly to the brain. While still in the development stage, it's one of the most promising technologies for regaining sight.



△ Seeing with your tongue

The BrainPort comes in two parts. The first is a pair of sunglasses with a digital camera. The second is a chip with 400 electrodes that sits on the user's tongue and translates the camera's video into electric signals. At first the data is meaningless, but with training, users can interpret visual images.